# Roots of polynomial functions

1 Determine whether the following values are zeros of the function $f\left(x\right)=x^4-7x^3+12x^2+4x-16$:

a $x=5$

b $x=4$

c $x=-1$

d $x=1$

2 For each of the following functions:

i State the zeros of the function.

ii State the multiplicity of each zero.

a $f\left(x\right)=\left(x+6\right)^3\left(x-3\right)\left(x+9\right)$

b $f\left(x\right)=\left(x+1\right)^2\left(x-6\right)\left(x+2\right)^5$

c $f\left(x\right)=-3\left(x-6\right)^4\left(x+2\right)^5x^3$

d $f\left(x\right)=x^3-3x^2-5x+15$

3 Determine whether each of the following equations have a root of multiplicity $4$:

a $\left(x+8\right)^4\left(x+3\right)=0$

b $\left(x+8\right)\left(x+3\right)^3=0$

c $\left(x+8\right)\left(x+3\right)=0$

d $\left(x+8\right)^3\left(x+3\right)=0$

4 Determine whether each of the following equations have a root of multiplicity $3$:

a $\left(x+1\right)\left(x^2+2x+1\right)\left(x+3\right)=0$

b $\left(x+1\right)^3\left(x+3\right)=0$

c $\left(x+1\right)^2\left(x+1\right)\left(x+3\right)=0$

d $\left(x+1\right)\left(x-9\right)\left(x+3\right)=0$

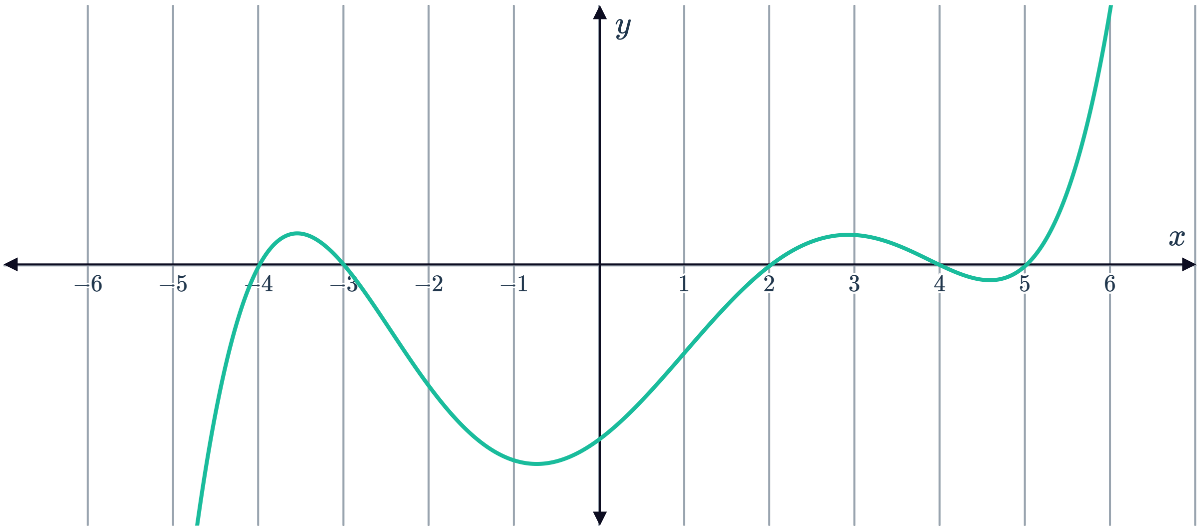
e $\left(x+2\right)\left(x-8\right)^3\left(x-1\right)=0$

f $\left(x^3+1\right)\left(x-2\right)\left(x+1\right)=0$

g $\left(x+3\right)\left(x-1\right)^2\left(x-3\right)=0$

h $\left(x^2-2x+1\right)\left(x-1\right)\left(x+9\right)^3=0$

**5** Consider the following graph of a function $f$.




**a** State all the linear factors of $f(x)$.

**b** Write an expression for a polynomial $h(x)$ with a leading coefficient of $1$, as a product of linear factors that has exactly the same roots as $f(x)$.

**c** Determine whether the following has roots which include all the roots of $f(x)$:

**i** $2(x+4)(x-5)(x+3)(x-4)(x-2)$

**ii** $(x+4)(x+3)(x-2)^2(x-5)$

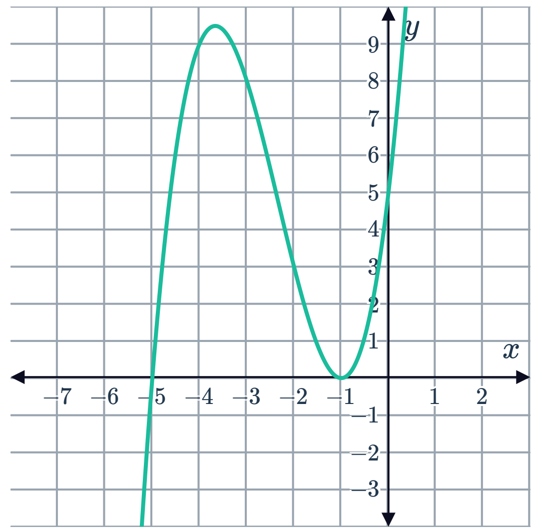
**iii** $4(x-4)(x-3)(x+2)(x+4)(x+5)$

**iv** $(x+4)(x+3)(x-2)(x-4)(x-5)(x+7)(x-11)$

**d** Apart from $f(x)$, how many other polynomials are there with the same roots and the same graph as $f(x)$?

## Equations of polynomial functions

**6** Consider the following graph of a cubic function:

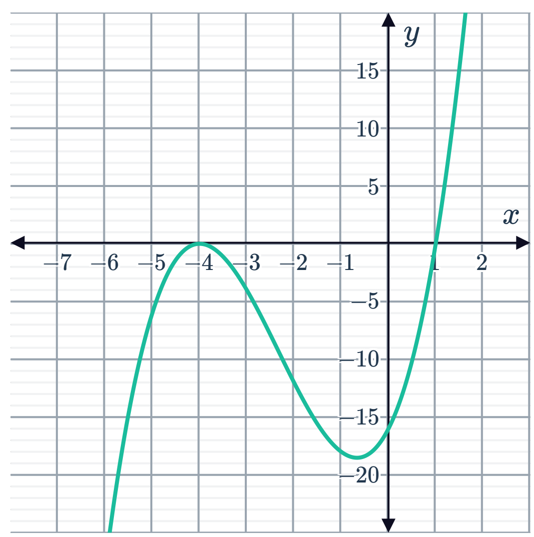



**a** State the $x$-intercepts of the function.

**b** State the $x$-intercept that has a double root.

**c** State the equation of the cubic function.

**7** Find the equation of the graphed cubic function:



**8** Find the equation of the following cubic fun[redacted]s in standard form:

**a**

- Has real coefficients
- Zeros at $x = 5$, $x = -3$, $x = 1$
- When $x = -2$, $y = -84$.

**b**

- Has real coefficients
- Zeros at $x = 0$, $x = 2$, $x = 5$
- When $x = -1$, $y = 72$.

**c**

- Has real coefficients
- Zeros at $x = 0$, $x = 6$, $x = -6$
- When $x = -4$, $y = 20$.

**d**

- Has real coefficients
- Zeros at $x = -5$, $x = 4$, $x = -1$
- When $x = 2$, $y = -126$.

**e**

- Has real coefficients
- Zeros at $x = 4$, $x = 1$, $x = 5$
- When $x = 2$, $y = 18$.

**f**

- Has real coefficients
- Zeros at $x = -1$, $x = -4$, $x = 3$
- When $x = -5$, $y = 64$.

**g**

- Has real coefficients
- Zeros at $x = -3$, $x = -1$, $x = -4$
- When $x = -2$, $y = -10$.

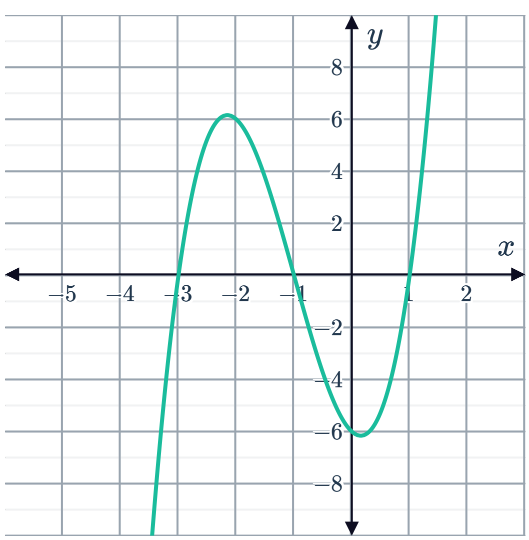
**h**

- Has real coefficients
- Zeros at $x = 4$, $x = 1$, $x = -2$
- When $x = 3$, $y = 50$.

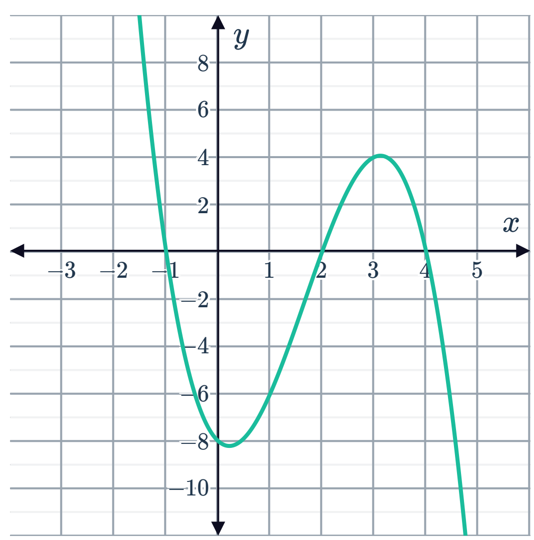
**9** Evan graphed a cubic function, $f(x) = ax^3 + bx^2 + cx + d$, and determined the roots of $f(x)$ to be $\pm 4$ and $5$. Find the value of $b$ if $a = 1$.

**10** Find the equation of the following graphed functions in factored form:

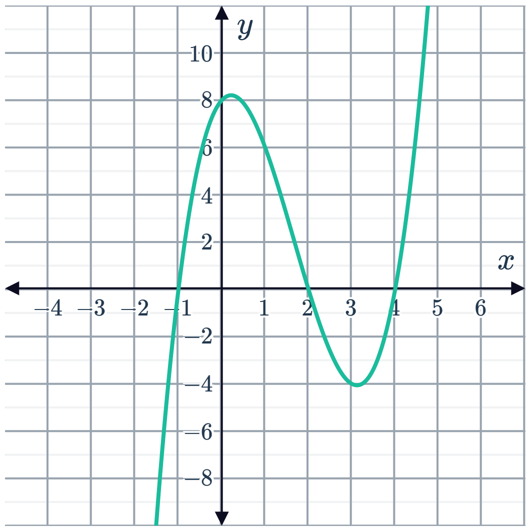
**a**

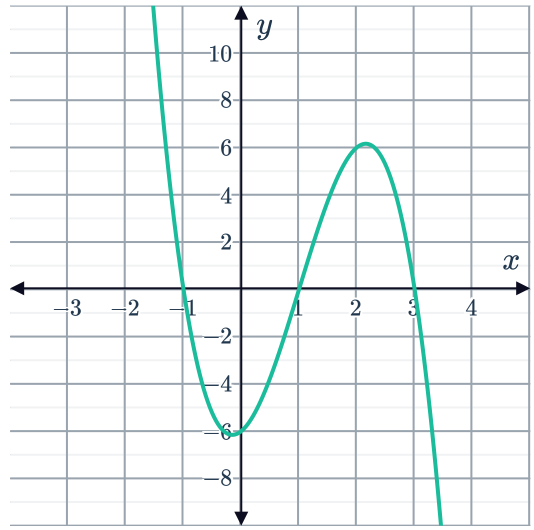



**b**

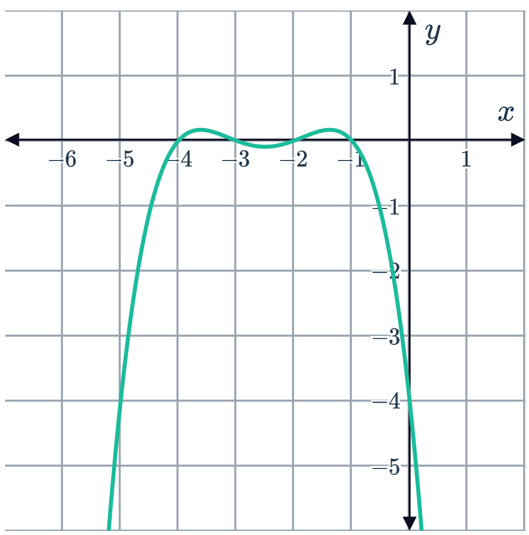


c

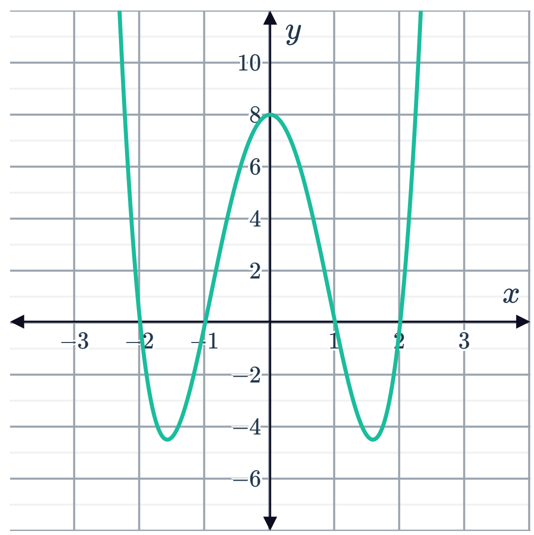

y
x
10
8
6
4
2
−2
−4
−6
−8
−4 −3 −2 −1 1 2 3 4 5 6


y
x
10
8
6
4
2
−2
−4
−6
−8
−3 −2 −1 1 2 3 4

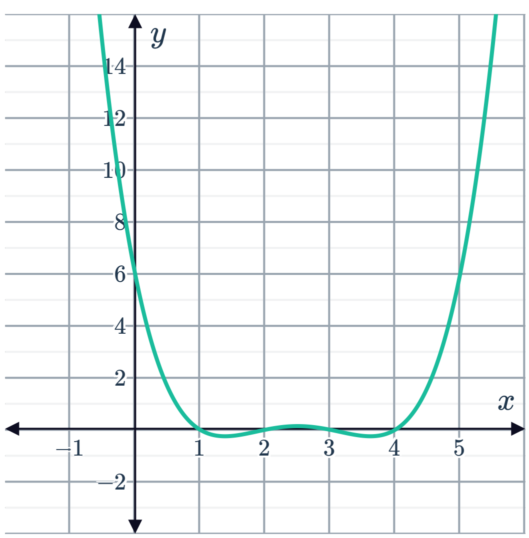
e


y
x
1
−1
−2
−3
−4
−5
−6 −5 −4 −3 −2 −1 1

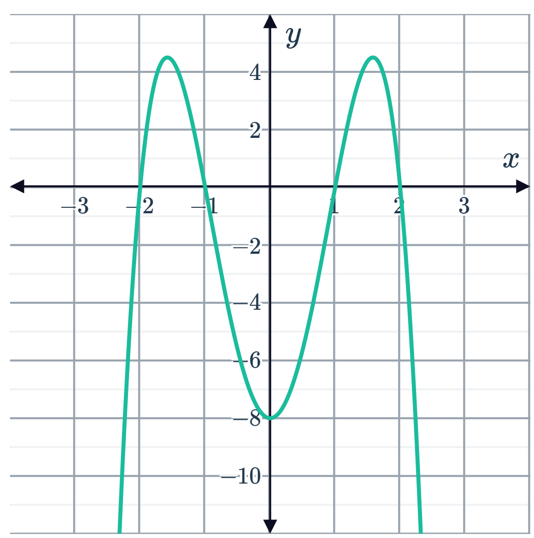
f


y
x
10
8
6
4
2
−2
−4
−6
−3 −2 −1 1 2 3

g


y
x
14
12
10
8
6
4
2
−2
−1 1 2 3 4 5

h


y
x
4
2
−2
−4
−6
−8
−10
−3 −2 −1 1 2 3